

Web Development Bootcamp: Week 2 Lesson Notes

Week 2: Cascading Style Sheets (CSS) Foundations & Layouts

Day 5: Introduction to Styling: Selectors, Fonts, and Colors

- **The Document Bridge:** HTML pages are completely blind to external styling blocks by default. You must explicitly patch an interconnection channel into the structural document head using:

```
<link rel="stylesheet" href="style.css">
```

- **CSS Rule Schema Structure** (*a clear property block syntax rule*):

```
selector {  
    property: value;  
}
```

- **Selectors:** The target identification filter defining precisely which HTML node element tree elements to manipulate (e.g., matching element types, class names, or specific IDs).
- **Properties:** The precise physical visual parameters you are commanding the engine to change (e.g., text color scales, font tracking families, backing elements layout metrics).
- **Values:** The configuration settings injected into that trait lane (e.g., setting background colors to absolute color space definitions or text to scaling fonts).

```
/* CSS Reset & General Font Rules */  
body {  
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;  
    background-color: #f8fafc; /* Soft slate white */  
    color: #1e293b; /* Off-black for crisp readability */  
}
```

```
/* Tag Selector Specific Overrides */
h1 {
  color: #4f46e5; /* Deep Indigo Theme Color */
  font-size: 2.5rem; /* Larger bold sizing display */
}

h2 {
  color: #0f172a; /* Midnight Blue */
  font-size: 1.75rem;
}

/* Target Custom Layout Text Paragraph Classes */
p {
  line-height: 1.6; /* Add comfortable vertical line spacing */
  color: #475569; /* Cool medium charcoal gray text color */
}
```

Day 6: The Box Model and Spacing

Core Concept: *The Four-Layer CSS Box Model: Content, Padding, Border, Margin.*

Content: The core internal text values or image dimensions.

Padding: space inside the card border layout lines.

Border: The outer container walls of an element.

Margin: The space between one element and another.

```

/* Turn Section Containers into Beautifully Spaced Content Cards */
section {
    background-color: #ffffff; /* Pure white card canvas fields */
    border: 1px solid #e2e8f0; /* Ultra-thin subtle gray layout card
border */
    border-radius: 12px; /* Smooth rounded interface component design
corners */

    /* Spacing Layout Architecture Implementation */
    padding: 30px; /* Internal breathing cushioning insulation room on
all 4 sides */
    margin-top: 20px; /* Push this card away from elements above it */
    margin-bottom: 20px; /* Push subsequent down away down below it */

    max-width: 800px; /* Prevent the content box wrapper from stretching
too wide */
}

```

Day 7: Flexbox Basics

Core Concept: *The Flexbox Axis Matrix: Parent Display Flex, Justify-Content, and Align-Items Layouts.*

display: flex; The main toggle button switch written on the parent container (like the `<ul>` navigation header link list container). It instantly turns nested child elements from stubborn static block pieces into responsive, flexible objects tracking horizontally next to each other.

justify-content: space-between; Tells child items: 'Spread out as far as possible! Pushing the first element to the left edge wall and sending the last node to the absolute right wall edge.'

align-items: center; The vertical level setter utility line. It automatically snaps elements of variable heights to match centered perfectly on a straight horizontal midpoint horizon baseline.

```

/* Target Header Navigation Layout Container Elements */
header nav ul {
    display: flex; /* Activate the flex magnet alignment field instantly */
    list-style: none; /* Strip off the ugly old black bullet list points */
    padding: 0;
    margin: 0;
}

/* Add custom layout spacing separation buffers between link list items */
header nav ul li {
    margin-right: 20px; /* Space items evenly horizontally */
}

/* Convert the primary Master Hero section element into a 2-Column Split view
*/
header {
    display: flex;
    justify-content: space-between; /* Move logo text left, menu items far
right */
    align-items: center; /* Center items on a clean horizontal timeline */
    background-color: #ffffff;
    padding: 15px 30px;
    border-bottom: 1px solid #e2e8f0;
}

```

Day 8: Styling Modern Interactive Elements & Components

Core Concept: *Pseudo-classes (:hover), Component States, Linear Transitions, and UI Inputs.*

- **The Base Component (e.g., button):** How an item registers by default when no cursor is near it.
- **The :hover State Modifier:** An invisible reactive tripwire hook. It says: 'The second a user's mouse cursor moves over this element boundary, overwrite the old styles and pop out these glowing colors instead!'
- **transition: 0.3s ease;** The slow-motion controller switch. Without this, colors snap instantly like a broken lightbulb. Adding transitions blends visual states smoothly over time, creating fluid animation effects.
- **UI Fields Styling:** Customizing input borders to glow neon blue when active (:focus), warning users clearly that a field is editing data live.

```
/* Build a Custom Action Link Button Component */
a.btn, button {
  display: inline-block;
  background-color: #4f46e5; /* Main Brand Indigo Indigo */
  color: #ffffff; /* Contrast white font text overlay */
  padding: 12px 24px;
  border-radius: 8px; /* Clean slick modern corners */
  text-decoration: none; /* Drop off the basic old underline link markers */
  font-weight: 600;
  border: none;
  cursor: pointer;

  /* Animation Blend Timing Controller Rule */
  transition: background-color 0.3s ease, transform 0.2s ease;
}

/* Interactive Hover Proximity State Adjustment Hook */
a.btn:hover, button:hover {
  background-color: #4338ca; /* Deepen color tone slightly on cursor overlap */
  transform: translateY(-2px); /* Give a tiny subtle visual floating pop lift lift */
}

/* Design Interactive Data Input Form Selection Fields */
input[type="text"], input[type="email"], textarea {
  width: 100%;
  padding: 12px;
  border: 1px solid #cbd5e1;
  border-radius: 6px;
  background-color: #ffffff;
  font-family: inherit;
  transition: border-color 0.2s ease;
}

/* Focus Input Validation Color Glow Adjuster */
input:focus, textarea:focus {
  outline: none;
  border-color: #4f46e5; /* Glow brand blue when typing fields actively */
  box-shadow: 0 0 0 3px rgba(79, 70, 229, 0.1);
}
```

